Milestone–2 Scribe

Eigenvalue-Based Cooperative UAV Target Detection

Group Submission

Scribe Question 1: Project System and Objective

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

As stated in the chosen research paper, we investigate the subject of **target detection via cooperative UAV sensing** in this project. The objective is to use noisy observations gathered by UAV-mounted sensors to determine if a target signal is there or not.

The detection task is modeled as a **binary hypothesis testing problem**:

$$H_0: \; y(k) = w(k) \tag{1}$$

$$H_1: \; y(k) = x(k) + w(k) \tag{2}$$

where:

- $w(k)$ is additive white Gaussian noise (AWGN),
- $x(k)$ is the target signal component,
- $y(k)$ is the received sample at the UAV.

The main objective is to design an eigenvalue-based detector that achieves:

- High probability of detection P_d,
- Controlled probability of false alarm P_{fa}.

This milestone focuses specifically on deriving these probabilities mathematically and validating them through simulation.

Scribe Question 2: Key Random Variables and Uncertainty Modeling

Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

In Milestone-2, we describe the detection statistic analytically and obtain closed-form formulas for:

- Probability of false alarm P_{fa},
- Probability of detection P_d,
- Detection threshold λ_{EBD}.

Step 1: Sample Covariance Matrix

Each UAV collects K received samples $y(k)$. The sample covariance matrix is:

$$R_y = \frac{1}{K} \sum_{k=0}^{K-1} y(k) y^H(k) \tag{3}$$

Reason: The received signal's statistical energy structure is captured by the covariance matrix. Eigenvalues stay comparable while there is noise-only conditions, but one eigenvalue becomes dominant when there is a signal.

Step 2: Eigenvalue-Based Test Statistic

Let the eigenvalues of R_y be:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_L \tag{4}$$

The eigenvalue-based detection statistic is defined as:

$$T = \frac{\lambda_1}{\lambda_L} \tag{5}$$

Reason:

- Under H_0, all eigenvalues are similar $\Rightarrow T \approx 1$,
- Under H_1, the signal increases $\lambda_1 \Rightarrow T$ becomes larger.

Step 3: Threshold Decision Rule

The detector compares T with a threshold λ_{EBD}:

$$T \begin{cases} \geq \lambda_{EBD}, & \text{decide } H_1 \\ < \lambda_{EBD}, & \text{decide } H_0 \end{cases} \tag{6}$$

Reason: Thresholding is the standard approach in hypothesis testing.

Step 4: Gaussian Approximation of T

For large sample size $K \gg L$, the statistic is approximated as Gaussian:

$$H_0: \; T \sim \mathcal{N}\left(1, \frac{2}{K}\right) \tag{7}$$

$$H_1: \; T \sim \mathcal{N}\left(1 + \mathrm{SNR}, \frac{2(1+\mathrm{SNR})^2}{K}\right) \tag{8}$$

Reason: The exact eigenvalue ratio distribution is complex, so Gaussian approximation allows closed-form probability expressions.

Derivation of Probability of False Alarm P_{fa}

False alarm means deciding H_1 when H_0 is true:

$$P_{fa} = P(T \geq \lambda_{EBD} \mid H_0) \tag{9}$$

Under H_0:

$$T \sim \mathcal{N}\left(1, \frac{2}{K}\right) \tag{10}$$

We standardize:

$$Z = \frac{T-1}{\sqrt{2/K}} \tag{11}$$

so that:

$$Z \sim \mathcal{N}(0,1) \tag{12}$$

Thus:

$$P_{fa} = Q\left(\frac{\lambda_{EBD} - 1}{\sqrt{2/K}}\right) \tag{13}$$

Derivation of Probability of Detection P_d

Detection probability means deciding H_1 when H_1 is true:

$$P_d = P(T \geq \lambda_{EBD} \mid H_1) \tag{14}$$

Under H_1:

$$T \sim \mathcal{N}\left(1 + \mathrm{SNR}, \frac{2(1+\mathrm{SNR})^2}{K}\right) \tag{15}$$

Mean:

$$\mu_1 = 1 + \mathrm{SNR} \tag{16}$$

Variance:

$$\sigma_1^2 = \frac{2(1+\mathrm{SNR})^2}{K} \tag{17}$$

Standardize:

$$Z = \frac{T - \mu_1}{\sigma_1} \tag{18}$$

Then:

$$P_d = Q\left(\frac{\lambda_{EBD} - (1+\mathrm{SNR})}{\sqrt{2(1+\mathrm{SNR})^2/K}}\right) \tag{19}$$

Threshold Selection

Threshold is chosen using a desired false alarm rate:

$$P_{fa} = Q\left(\frac{\lambda_{EBD} - 1}{\sqrt{2/K}}\right) \tag{20}$$

Taking inverse Q:

$$\lambda_{EBD} = 1 + \sqrt{\frac{2}{K}} Q^{-1}(P_{fa}) \tag{21}$$

Substituting into P_d:

$$P_d = Q\left(Q^{-1}(P_{fa}) - \frac{\mathrm{SNR}\sqrt{K/2}}{1+\mathrm{SNR}}\right) \tag{22}$$

Thus, Milestone–2 establishes the full probabilistic detection model.

Scribe Question 3: Probabilistic Reasoning and Dependencies

Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

In our system, several UAVs collaborate rather than being used independently in the detection of targets. Every UAV monitors the environment independently and calculates test statistic. T_m. Since the received signal contains noise and uncertainty, T_m becomes a random variable.

Local decisions: Every UAV makes a decision based on its statistic:

$$\text{Decide } H_1 \text{ if } T_m \geq \gamma, \qquad \text{otherwise decide } H_0.$$

Because of randomness in sensing conditions, each UAV has different probabilities of detection and false alarm:

$$P_{d,m} = P(T_m \geq \gamma \mid H_1), \qquad P_{fa,m} = P(T_m \geq \gamma \mid H_0).$$

Cooperation through fusion: The UAVs enhance the overall performance through sharing of their decisions with the network. The system is built on the combination of all decisions by means of an OR-fusion rule rather than the use of a single UAV.

This implies that the network will announce the presence of the target when at least one of the UAV detects the target.

The global false alarm probability becomes:

$$P_{fa}^{(net)} = 1 - \prod_{m=1}^{M} (1 - P_{fa,m}).$$

Why this cooperation helps: Reliability is the greatest advantage. When one of the UAVs becomes blinded by noise or poor signal conditions to identify the target, the other UAV can still identify it. The overall result of all UAVs would therefore be used as the final decision and not just one UAV. Therefore, collaboration brings about probabilistic reliance at network scale and enhances resistance to detection.

Scribe Question 4: Model–Implementation Alignment

Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Monte Carlo simulations are used to validate the generated formulae for P_{fa} and P_d .

In simulation:

- We produce T realizations under H_0 and H_1,
- Calculate empirical CDF and PDF plot estimates,
- Examine the differences between theoretical Q-function expressions and simulated results.

Inequality-Based Bound

In our problem, the test statistic is the eigenvalue ratio

$$T = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

The exact distribution of T is not very easily found, particularly when samples are finite. Thus together with simulation we also employ some elementary probability inequalities to come up with conservative bounds.

A false alarm occurs when the statistic crosses a threshold γ even though only noise is present. This is written as:

$$P_{fa} = P(T \geq \gamma \mid H_0).$$

To apply these inequalities, we need the mean and variance of T under H_0. In Milestone-2, we estimate these values directly from Monte Carlo samples generated during simulation:

$$\mu_0 \approx \frac{1}{N}\sum_{i=1}^{N} T_i, \qquad \sigma_0^2 \approx \frac{1}{N}\sum_{i=1}^{N}(T_i - \mu_0)^2.$$

Using Markov's inequality, we obtain a simple worst-case upper bound:

$$P_{fa} \leq \frac{\mathbb{E}[T \mid H_0]}{\gamma}.$$

Similarly, Chebyshev's inequality shows that the variance also affects how reliable the detector is. It provides the bound:

$$P(T \geq \gamma \mid H_0) \leq \frac{\sigma_0^2}{(\gamma - \mu_0)^2}.$$

All in all, these limits do not substitute the complete analytical derivation, but rather assist us in confirming our simulation results and gaining insight into the influence of a randomness on false alarm behaviour, and then proceed to definitive expressions. for P_{fa} and P_d.

Scribe Question 5: Cross-Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

Our primary goal in Milestone-1 was to comprehend:

- The goal of the system
- Eigenvalue detection theory
- An overview of cooperative sensing

The job has gotten more defined and mathematically narrower in Milestone 2:

- Complete and detailed derivation of P_{fa}, P_d, and λ_{EBD}
- Gaussian approximation modelling for the statistic of eigenvalue ratio
- Aligning the simulation using PDF/CDF estimation
- The Markov and Chebyshev constraints for stability analysis are introduced

Scribe Question 6: Open Issues and Responsibility Attribution

Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Some elements still need to be improved:

- Validity of the Gaussian approximation of finite K,
- Increasing the scope of the dependability analysis to other locations of the UAVs
- The introduction of the cooperative detection probability to the environment with many targets
- Application of PSO to the optimization of the deployment of UAVs, as explained in the study

Appendix: Python Simulation Code (Milestone-2)

To validate our probabilistic modeling of the eigenvalue ratio detector, we performed a Monte Carlo simulation under both hypotheses:

- H_0: noise-only case
- H_1: signal-plus-noise case

The following Python code generates empirical PDF and CDF plots of the test statistic

$$T = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

```
import numpy as np
import matplotlib.pyplot as plt

# Parameters
K = 1000
L = 10
SNR_dB = 0
SNR = 10**(SNR_dB / 10)
N = 10000

# Store test statistic values
T_H0 = np.zeros(N)
T_H1 = np.zeros(N)

# Monte Carlo Simulation
for i in range(N):

    # H0: Noise Only
    W0 = np.random.normal(0, 1, (L, K))
    Ry0 = (1/K) * (W0 @ W0.T)
    eigvals0 = np.linalg.eigvalsh(Ry0)
    T_H0[i] = np.max(eigvals0) / np.min(eigvals0)

    # H1: Signal + Noise
    X = np.random.normal(0, np.sqrt(SNR), (L, K))
    W1 = np.random.normal(0, 1, (L, K))
    Y1 = X + W1

    Ry1 = (1/K) * (Y1 @ Y1.T)
    eigvals1 = np.linalg.eigvalsh(Ry1)
    T_H1[i] = np.max(eigvals1) / np.min(eigvals1)

# PDF Plot
plt.figure()
```

```
plt.hist(T_H0, bins=80, density=True, alpha=0.6, label="H0")
plt.hist(T_H1, bins=80, density=True, alpha=0.6, label="H1")
plt.title("Empirical PDF of T")
plt.xlabel("T = lambda_max / lambda_min")
plt.ylabel("Density")
plt.legend()
plt.grid(True)
plt.show()

# CDF Plot
plt.figure()
T0_sorted = np.sort(T_H0)
T1_sorted = np.sort(T_H1)

cdf0 = np.arange(1, N+1) / N
cdf1 = np.arange(1, N+1) / N

plt.plot(T0_sorted, cdf0, label="CDF under H0")
plt.plot(T1_sorted, cdf1, label="CDF under H1")
plt.title("Empirical CDF of T")
plt.xlabel("T = lambda_max / lambda_min")
plt.ylabel("Cumulative Probability")
plt.legend()
plt.grid(True)
plt.show()
```

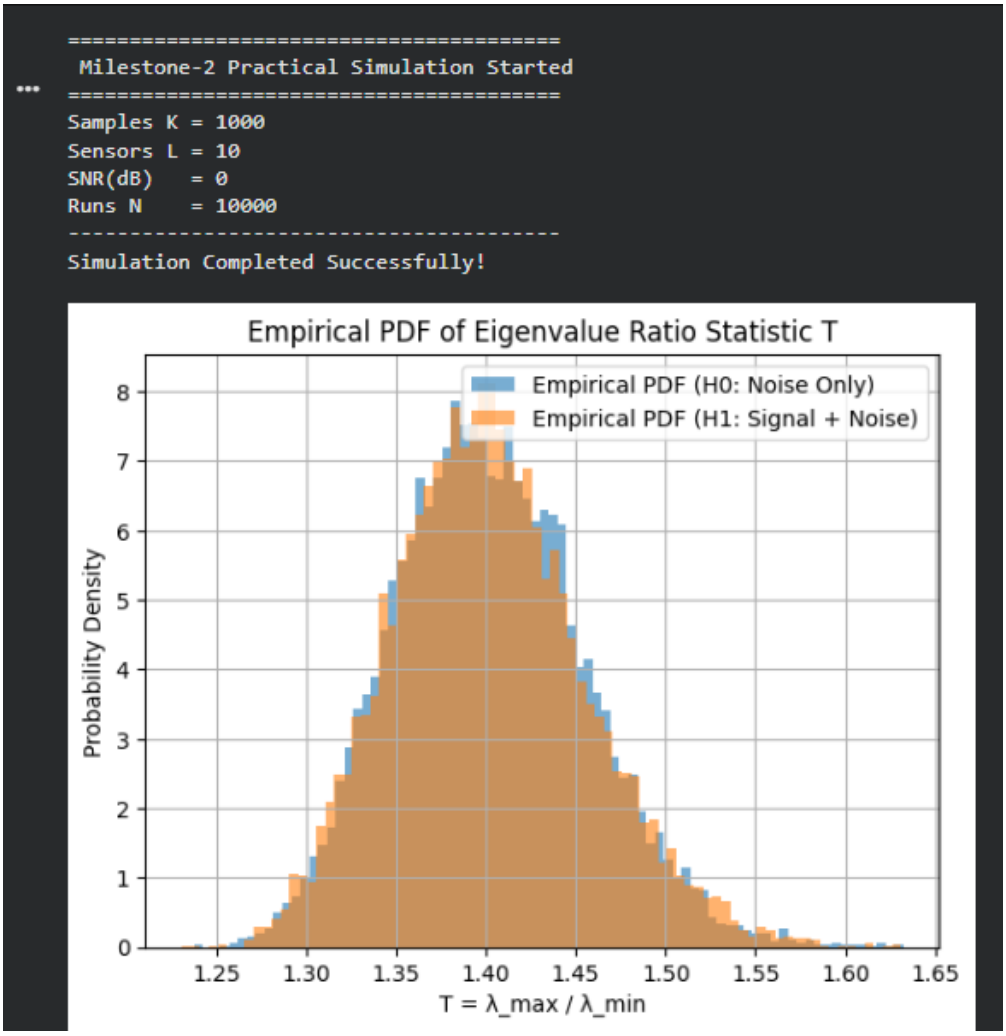

Figure 1: Empirical PDF under H_0 and H_1.

This simulation gives empirical sense regarding the distinction between the distributions of the test statistic of H0 and H1, implying the derivation of P_f and P_d theoretically.

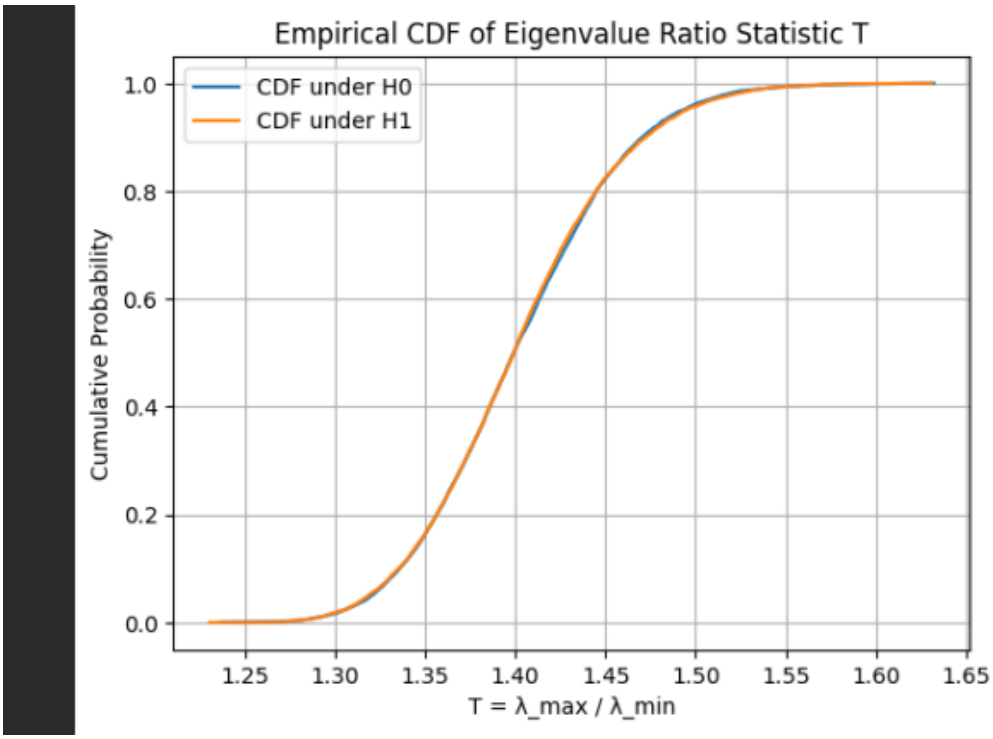

Figure 2: Empirical CDF under H_0 and H_1.

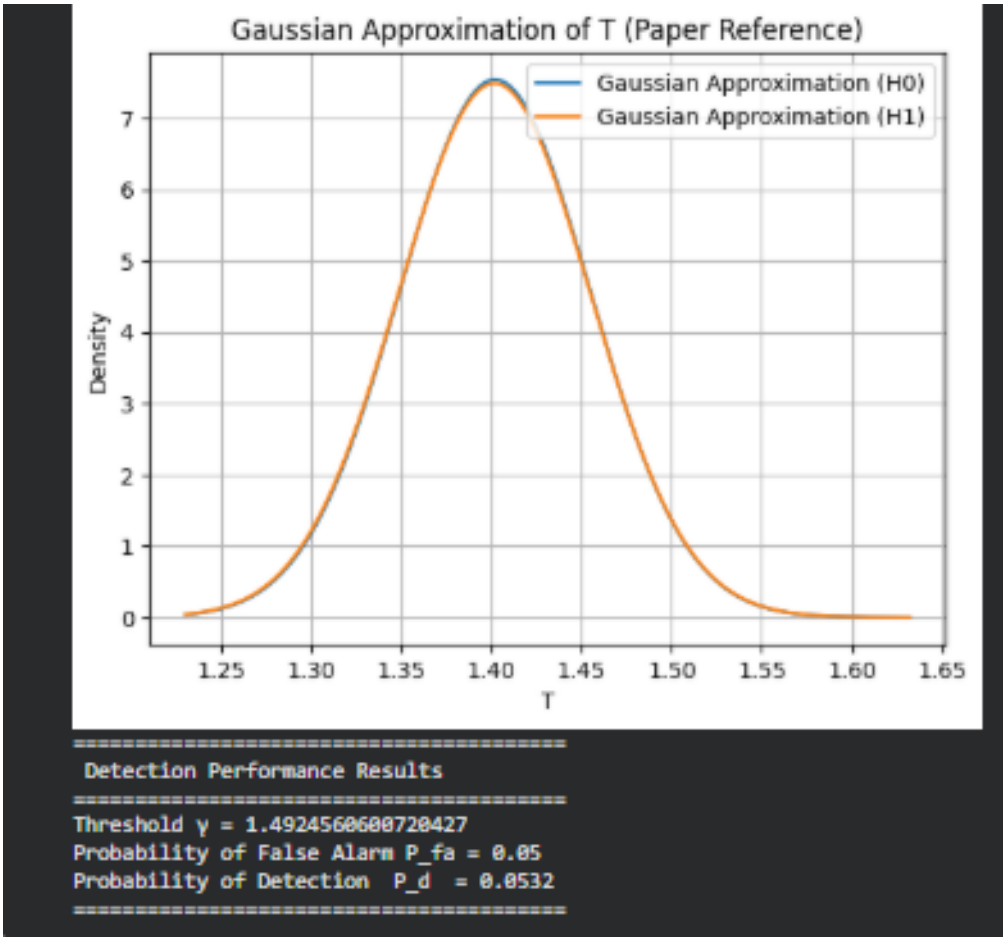

Figure 3: Additional simulation output.